

1 MAP from Gaussian Posterior

1.1 Input and Assumption Nodes

Input node: `posterior.gaussian.conjugate`.

1.2 Derivation

For a Gaussian posterior $p(\beta \mid X, y) = \mathcal{N}(\mu_n, \Sigma_n)$, the density is maximized at its mean:

$$\hat{\beta}_{MAP} = \mu_n.$$

1.3 Result

The MAP estimator equals the posterior mean in `estimator.beta_map`.